

Metric-Aware Latent Diffusion via Learnable Riemannian Geometry

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- Motivation & Related Works
- Methodology
- Numerical Results



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We study latent Diffusion Models from two perspectives:

1. *Can we preserve the intrinsic geometry of the data in the latent manifold?*
 - ▶ An encoder preserving the geometry of the data space.
2. *Can we merely make a minimal assumption about the latent manifold?*
 - ▶ No prior assumption on the latent manifold except for the smoothness of the manifold.

Problem

Can we preserve the *intrinsic geometry* of the data in the latent manifold?

- ▶ Traditional methods like PCA and Autoencoders (AEs) often distort intrinsic manifolds due to the lack of geometric constraints.
- ▶ *Isometry* is a natural idea and has been widely applied [RS00; MHM18; MZ24].
- ▶ To address this, recent research has imposed isometry via regularization [GAL20; Lee+22].
- ▶ However, they assume the flat latent manifold.

Problem

Can we merely make a *minimal assumption* about the latent manifold?

- ▶ [Hua+22; De +22; ZGK23] generalize diffusion to curved spaces.
- ▶ However, these methods depend on *pre-defined manifold structures*.
- ▶ Standard latent diffusion models [VKK21; Lee+23] lack geometric interpretability.



We explore the *metric-aware* latent diffusion model.

1. Learn a latent Symmetric Positive Definite (SPD) metric directly from the data.
2. Align it with the decoder's pull-back geometry.
3. Run score-based diffusion *on* that learned manifold.

Remark

*These bridge the gap between *geometry-aware autoencoders* and *manifold-aware diffusion*.*



- Motivation & Related Works
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1. **Data manifold.** Observations $\mathcal{D} = \{x_i\}_{i=1}^N \subset \mathbb{R}^D$ lie near an unknown m -dimensional manifold $\mathcal{M}$ ($m \ll D$).
2. **Autoencoder.** Encoder $g_\phi : \mathbb{R}^D \rightarrow \mathbb{R}^m$ and decoder $f_\theta : \mathbb{R}^m \rightarrow \mathbb{R}^D$ map $x \mapsto z \mapsto \hat{x}$ with latent $z = g_\phi(x)$.
3. **Latent geometry.** Equip each latent point with a learnable SPD metric $G_\psi(z) \in \mathbb{R}^{m \times m}$, making $(\mathcal{M}, G_\psi)$ a Riemannian manifold.
4. **Generative objective.** Learn a score network $s_\varphi(z, t)$ such that running the reverse SDE on $(\mathcal{M}, G_\psi)$ produces samples whose push-forward distribution matches the data distribution $p_{\text{data}}(x)$.

- ▶ Let $f : \mathcal{M} \rightarrow \mathcal{N}$ be a smooth mapping, represented in local coordinates by $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$.
- ▶ Its differential is denoted by the Jacobian matrix $J_f(z) := \frac{\partial f}{\partial z}(z) \in \mathbb{R}^{n \times m}$. $J_f(z)$ actually pushes a tangent vector on $T_z \mathcal{M}$ to the one on $T_{f(z)} \mathcal{N}$.
- ▶ f is isometry if the mapping preserves distances and angles for any $z \in \mathbb{R}^m$. That is,

$$G(z) = J_f(z)^\top H(f(z)) J_f(z), \quad \forall z \in \mathbb{R}^m. \quad (\text{Pullback Metric})$$

Remark

The isometry can be verified by the above equation.

- ▶ Setting $H(f(z))$ as the identity, we arrive at the pull-back metric

$$G(z) = J_f(z)^\top J_f(z), \quad \forall z \in \mathbb{R}^m.$$

- ▶ **Metric parameterisation:** We want to find the **Metric Family** in the latent, which is smooth by construction and guarantees SPD: $G^{-1}(z) = \sum_{i=1}^M L_i L_i^\top e^{-\frac{\|z - c_i\|^2}{T^2}} + \lambda I$.
 - ▶ L_i, c_i, T are a trainable Cholesky factor, kernel centers, a kernel bandwidth, respectively.
 - ▶ This blends low-rank factors with Gaussian RBF's.
- ▶ **Isometry loss:** We then have a training loss that uses a pull-back metric $G_{\text{pb}} = J^\top(z)J(z)$ w.r.t. the Frobenius norm coupled with a curvature variance regularizer:

$$\mathcal{L}(\psi) = \mathbb{E}_{z \sim q_\phi} [\|\tilde{G}_\psi - \tilde{G}_{\text{pb}}(z)\|_F^2] + \lambda_{\text{curv}} \text{Var}_{z \sim q_\phi} [\kappa_{\text{sec}}(z)]$$

where the expectation is w.r.t. the aggregate posterior of the VAE.

- ▶ For the curvature regularizer, the squared Frobenius norm of the Jacobian of G is computed.
 - ▶ In practice, Hutchinson estimator is used over a mini-batch to get an unbiased trace estimate.
- ▶ **Conditioning regularizer:** $\mathcal{R}_{\text{cond}} = \mathbb{E}[\|\log \text{diag}(L_\psi)\|_2^2]$ keeps eigenvalues in range.

With G in hand, we want the shortest path (*geodesics*) between two latent codes $z_0, z_1 \in \mathbb{R}^m$.

- ▶ Given a smooth curve $\gamma: [0, 1] \rightarrow \mathcal{M}$, it has an *energy* of

$$E[\gamma] = \frac{1}{2} \int_0^1 \dot{\gamma}(t)^\top G(\gamma(t)) \dot{\gamma}(t) dt, \quad \dot{\gamma}(t) = \frac{d\gamma}{dt}.$$

- ▶ Geodesics are the curves that minimise E subject to $\gamma(0) = z_0, \gamma(1) = z_1$.¹

Remark

*If the trained G is close to the true metric, the decoded geodesic should *lie on* $\mathcal{M}$.*

¹The length is $L[\gamma] = \int \sqrt{2E[\gamma]} dt$ which minimizes E and L equivalently.



The diffusion in the latent space is as follows:

- ▶ **Forward SDE (flattened):** $dy_t = \sqrt{2} dw_t$, $z_t = L_\psi^{-\top}(y_t)$.
- ▶ **Score loss:** $\mathcal{J}(\varphi) = \mathbb{E}[\|s_\varphi - \sqrt{2}\sigma_t^{-1}L_\psi^{-\top}w\|_{G_\psi}^2]$, reduces to MSE in y -space.
- ▶ **Reverse SDE:** Following [Hua+22], the time-reverse SDE on z_t is

$$dz_t = [2 \operatorname{div}(G_\psi^{-1}) - s_\varphi(z_t, t)]dt + \sqrt{2} L_\psi(z_t)^{-\top} d\bar{w}_t, \quad (1)$$

Hutchinson probe supplies it for free.

For the training:

1. **VAE pre-train** (ϕ, θ) :
2. **Metric network train** (ψ) : freeze (ϕ, θ) , train ψ .
3. **Score network train** (φ) : freeze (ϕ, θ, ψ) , and train φ .



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VAE fails to reconstruct the Swiss-roll data.

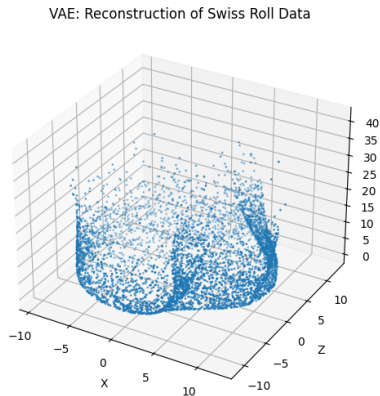
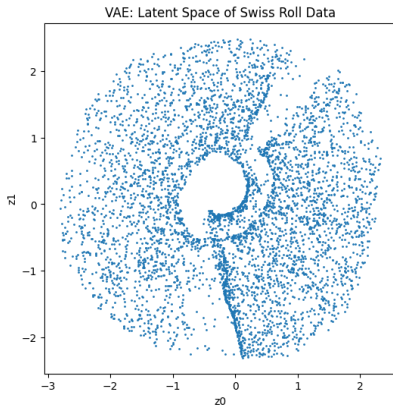


Figure: Visualization of the latent space and the generated samples from VAE

Even with the isometry, ISO-VAE fails to reconstruct the Swiss-roll data.

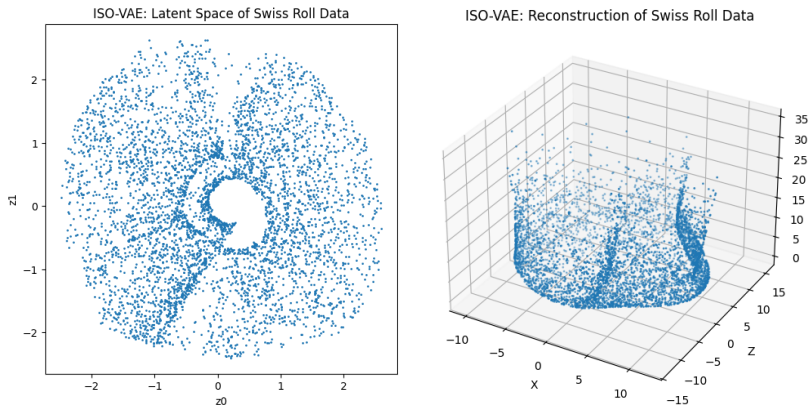


Figure: Visualization of the latent space and the generated samples from ISO-VAE

One can verify the validity of the learned latent metric by observing the geodesic.

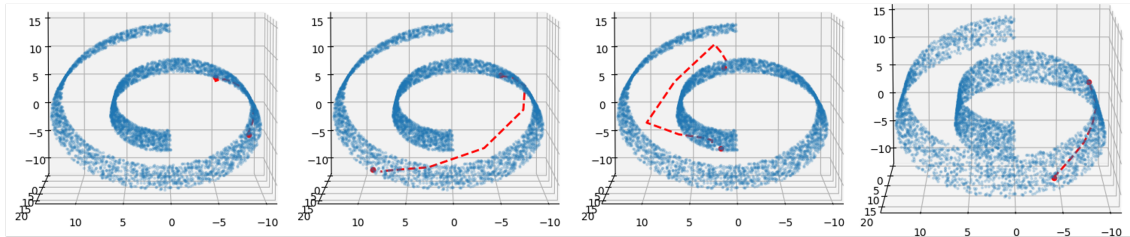


Figure: Visualization of Swiss-roll (scatter) with decoded geodesic (red line)

Score-based SDE on the learned latent manifolds



The proposed method is capable of generating the data from Swiss-roll manifold.

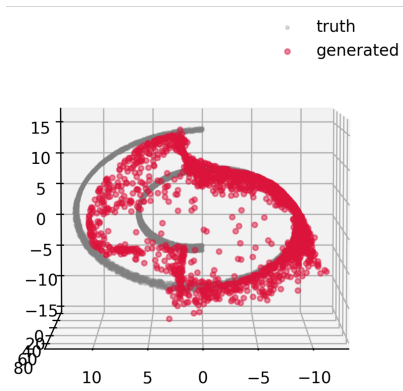


Figure: Generated points on Swiss-roll manifold using the Score-based SDE. The red scatter points represent the generated data.



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